Reference: McGuinness, D. L. (2004). "Ontology Development 101"

Summary of the Paper

McGuinness provides a foundational methodology for building ontologies, emphasizing best practices for modeling domains. It covers core principles such as determining the scope of the ontology, defining hierarchical class structures, and establishing logical relationships (properties) to enable semantic reasoning.

Application in Our Project:

- Methodological Foundation: We adopted the principles from "Ontology Development 101" to ensure our Harry Potter Knowledge Graph was built on a rigorous, academic framework rather than an ad-hoc structure.
- Hierarchical Integrity: Following these guidelines, we meticulously defined our class hierarchy (e.g., Student, Professor, and Dark Wizard as subclasses of Person) to maximize the efficiency of automated reasoners.
- Logical Consistency: We implemented property characteristics—such as symmetric and inverse properties—as recommended by McGuinness to ensure our ontology remained consistent and support complex inferential queries.

Reference: Yani, M., & Krisnadhi, A. A. (2021). "Challenges, Techniques, and Trends of Simple KGQA: A Survey"

Summary of the Paper

This survey provides a comprehensive review of the state-of-the-art in Knowledge Graph Question Answering (KGQA) systems. It categorizes the diverse techniques used to bridge natural language queries with structured semantic data and outlines the primary challenges in maintaining accuracy and scalability.

Application in Our Project

- Methodological Validation: We utilized the evaluation frameworks discussed in this survey to design our set of 22 competency questions, ensuring our testing methodology aligns with standard KGQA assessment practices.
- Technical Context: This paper informed our approach to the "NL-to-SPARQL" module by highlighting the effectiveness of integrating Large Language Models (LLMs) as a translation layer.

- Future-Proofing: Our "Limitations & Future Work" section was guided by the trends identified in this survey, specifically concerning the need for more robust, user-friendly natural language interfaces and the integration of external knowledge sources.

Reference: D'Abramo, J., et al. (2025). "Investigating large language models for text-to-sparql generation" (بخش خودمه)

Summary of the Paper

This study, presented at the 4th International Workshop on Knowledge-Augmented Methods for NLP, examines the capability of Large Language Models (LLMs) to accurately translate natural language queries into executable SPARQL code. It explores the challenges of schema-aware generation and demonstrates the effectiveness of prompting techniques in bridging the gap between human language and structured knowledge graphs.

Application in Our Project

- Architectural Innovation: This paper provided the architectural framework for our "NL-to-SPARQL" module, validating our use of Few-Shot Learning to map natural language to our specific Harry Potter ontology schema.
- Technical Justification: We adopted the techniques discussed in this research to ensure our system could handle complex semantic queries, such as professor-spell mappings and character relationship lookups.
- State-of-the-art Alignment: Citing this 2025 research demonstrates that our project implements current, cutting-edge methodologies for knowledge-augmented natural language processing.